

Finite Fields

A finite field F , written as $\{F, +, \times\}$, is a finite set of elements with two binary operations called *addition* and *multiplication*, such that for all $a, b, c \in F$ the following axioms are obeyed:

$a + b \in F$	(closure under addition)
$a + (b + c) = (a + b) + c$	(associativity of addition)
$a + 0 = 0 + a = a$	(addition identity)
$a + (-a) = (-a) + a = 0$	(inverse for addition)
$a + b = b + a$	(commutativity of addition)
$ab \in F$	(closure under multiplication)
$a(bc) = (ab)c$	(associativity of multiplication)
$a(b + c) = ab + ac$	(distributive law)
$ab = ba$	(commutativity of multiplication)
$a1 = 1a = a$	(multiplication identity)
$ab = 0$ iff $a = 0$ or $b = 0$	(can't multiply two different non-zero elements and get a zero)
$aa^{-1} = a^{-1}a = 1, \quad a \neq 0$	(inverse for multiplication, where $a \neq 0$)

For a prime number p , the set of non-negative integers $\mathbb{Z}_p$ with addition and multiplication modulo p form a finite field. Because this finite field has p elements we denote it as $GF(p)$. The "GF" is short for Galois field; a finite field with order p .

Examples:

Show that $\mathbb{Z}_3$ with addition and multiplication modulo 3 is a finite field.

Show that $\mathbb{Z}_4$ with addition and multiplication modulo 4 is NOT a finite field.

Finite Fields with order 2^n

We showed that $\mathbb{Z}_4$ with addition and multiplication modulo 4 is NOT a finite field... but there is a Galois field of order 4, that is a finite field... but its elements are polynomials!

Addition mod 4

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Multiplication mod 4

.	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

Addition GF(4)?

+	0	1	2	3
0	0	1	2	3
1	1	0	3	2
2	2	3	0	1
3	3	2	1	0

Multiplication GF(4)?

.	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	3	1
3	0	3	1	2

Let $m(x)$ be an irreducible polynomial of degree n with coefficients $\{0,1,\dots,p-1\}$. Then there are p^n different polynomials in the set of all polynomials with coefficients $\{0,1,\dots,p-1\} \bmod m(x)$. These p^n polynomials form a finite field.

Example: Let $p=2$, $n=2$, and $m(x) = x^2 + x + 1$ be the irreducible polynomial of degree $n=2$.

Then there are $2^2 = 4$ polynomials in the set. They are: 0 , 1 , x , and $x+1$.

The **four changes** to addition are:

$$1+1=2 \equiv 0 \bmod 2$$

$$1+(x+1)=(x+2) \equiv x \bmod 2, \text{ and } (x+1)+1=(x+2) \equiv x \bmod 2$$

$$(x+1)+(x+1)=(2x+2) \equiv 0 \bmod 2$$

For **four changes** to multiplication are:

$$x(x) = (x^2) \equiv (x+1) \bmod m(x)$$

$$x(x+1) = (x^2+x) \equiv 1 \bmod m(x), \text{ and } (x+1)x = (x^2+x) \equiv 1 \bmod m(x)$$

$$(x+1)(x+1) = (x^2+2x+1) \equiv x \bmod m(x)$$

Now, using the polynomials $\{0, 1, x, x+1\}$ instead of the integers $\{0,1,2,3\}$ we can complete the addition and multiplication tables for $GF(4)$... which is usually written as $GF(2^2)$ to clearly show that $p=2, n=2$.

Addition $GF(2^2)$

+	0	1	x	x+1
0	0	1	x	x+1
1	1	0	x+1	x
x	x	x+1	0	1
x+1	x+1	x	1	0

Multiplication $GF(2^2)$

.	0	1	x	x+1
0	0	0	0	0
1	0	1	x	x+1
x	0	x	x+1	1
x+1	0	x+1	1	x

Representing the polynomials $\{0, 1, x, x+1\}$ using 2-bit binary:

$0 \rightarrow 00$

$1 \rightarrow 01$

$x \rightarrow 10$

$x+1 \rightarrow 11$

Addition $GF(2^2)$

+	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

Multiplication $GF(2^2)$

.	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

Exercises:

1. Show how each of the **six** representations shown in the $GF(2^2)$ addition table are derived by:
 - Converting the associated row/column into polynomials and adding
 - Reducing the result mod 2 and converting back to binary

+	00	01	10	11
00				
01		00		
10		11	00	
11		10	01	00

2. Show how each of the **six** representations shown in the $GF(2^2)$ times table are derived by:
 - Converting the associated row/column into polynomials and multiplying
 - Reducing the result mod $m(x) = x^2 + x + 1$ and converting back to binary

.	00	01	10	11
00				
01		01		
10		10	11	
11		11	01	10

3. Show that $\mathbb{Z}_5$ with addition and multiplication modulo 5 is a finite field.
4. Show that $\mathbb{Z}_6$ with addition and multiplication modulo 6 is NOT a finite field.
5. Are the following finite fields? Explain your answers.
 - a) $\mathbb{Z}_2$ with addition and multiplication modulo 2
 - b) $\mathbb{Z}_{13}$ with addition and multiplication modulo 13
 - c) $\mathbb{Z}_{26}$ with addition and multiplication modulo 26